

EU SAE Application Package

DOWNLOAD AND VERIFICATION INSTRUCTIONS

Version 5.2.0-rc.6-wizard.5.10

Release candidate. This package is a release candidate for review and testing. Whether to publish estimates as official statistics is for each national statistical office to decide. Download it only from the project's GitHub Releases page and never use GitHub's automatic "Source code" archives.

1. Obtain the correct archive

Open github.com/Noboyoshida227/eu-sae-application/releases and, under **Assets**, download **EU_SAE_520_w5j.zip**. Its top-level folder is **EU_SAE_520_w5j**. Do not use the automatic "Source code" archives and do not ZIP a working copy of the repository.

2. Verify the package

Open **VERSION** and confirm 5.2.0-rc.6; open **WIZARD_VERSION** and confirm 5.2.0-rc.6-wizard.5.10. From the extracted package root, run **Rscript scripts/verify_release.R** to check the full file set and SHA-256 hashes. Compare the ZIP's SHA-256 with the value on the release page (Windows: **Get-FileHash <zip> -Algorithm SHA256**; macOS: **shasum -a 256 <zip>**). Stop if a listed file is missing or changed, or if an unexpected file is present.

3. Check the expected contents

The clean package contains the application, R helpers, scripts, tests, output placeholders, instructions, guidance, licence/notice files, and the bundled **Data/simulated/** and **Data/Spain/** examples. Internal notes, .git, app_runs, user-created Data folders, local libraries, credentials, generated outputs, and local diagnostics must not ship. The Spain survey and auxiliary files retain their GPL-2 terms. The boundary uses documented IGN/CNIG CartoBase ANE under CC BY 4.0; see **Data/Spain/README.md** and **THIRD_PARTY_NOTICES.md**. The Spain survey derivation script remains missing.

4. Install and start

Install **R 4.2 or later**. Use the newest R version approved and available in your organization; the absolute latest R release is not required. On a managed computer, keep the approved R installation unless IT directs an upgrade. Run **install_packages.R** once with internet access.

HTML and Word reports need Pandoc: the launcher reuses one already installed (RStudio Desktop, Quarto, Positron or [standalone Pandoc](#)) and otherwise downloads a pinned, checksum-verified copy into your user profile on first run (26-42 MB, no administrator rights). Without it the analysis still runs and the report is skipped. On Windows, open **Start_Here** and double-click a .bat launcher. On macOS, double-click **Start_Wizard.command** or **Start_Dashboard.command**; the first launch may require Control-click, then Open. On Linux, run **bash Start_Here/Start_Wizard.command** or the Dashboard equivalent. Keep the terminal window open. See **Start_Here/README.md** for Mac Gatekeeper and executable-permission help.

5. Follow the wizard flow (or dashboard equivalent)

The wizard presents the same inputs as six steps: 1 Data, 2 Mapping, 3 Indicator, 4 Models, 5 AI Assistant (optional consent, key, and language), and 6 Review & Run. On the final step, use **1. Check Data Readiness** before **2. Run Analysis**. For each auxiliary covariate, Data Readiness reports its correlation with the target indicator with a p-value and significance codes, by year and pooled, most significant first; use it for screening only. Its tables are saved in **outputs/tables/** and kept with the run. The analysis runs in a separate R process: the page stays responsive, the log updates live, and **Stop** ends the run (files written so far stay in outputs/; nothing is archived). The breadcrumb is clickable: green means complete, amber means an outstanding item, and blue means the current step. Navigation is soft, but the run control stays disabled until fatal data requirements are met. **Load Last Setup** is available from every step; API keys and transfer consent are never persisted. The classic dashboard exposes the same controls in one sidebar.

Prepare and run an analysis

INPUTS, MODEL OPTIONS, REPRODUCIBILITY, AND OPTIONAL AI

6. Select approved local inputs

For learning, use either bundled example. **Data/simulated/** contains fully synthetic survey, auxiliary, and geometry files plus a seeded generator. **Data/Spain/** contains the guidance example with mapping, attribution, and licence details. Neither example may feed a real estimate. For actual analyses, browse to approved household survey microdata, domain-level auxiliary covariates, geometry, and optional population or benchmark-target files. Confirm provenance, permitted use, sensitivity classification, area codes, and year coverage before running. Results remain under **outputs/**; see **Data/README.md** for saved-setup compatibility.

7. Apply the revised variance rule

For UFH and MFH, **sm_out** replaces a direct sampling variance with its smoothed value only when the direct value is missing/non-finite or strictly below 0.001. A direct variance equal to 0.001 is retained. **sm_all** replaces all direct variances; **direct** retains direct values subject to documented safety backfills.

8. Choose the benchmarking level explicitly

After selecting Apply benchmarking, choose National or Grouped. National uses one countrywide group and constrains population-weighted domain estimates to the direct national target. Grouped requires a mapped higher-level survey variable, such as region or NUTS2, and applies the constraint within each group. An external Benchmark Target Database remains optional.

9. Record reproducibility and inference controls

The seed defaults to 123. MFH MCPE bootstrap uses **mcpe_nB** (interactive default 200; minimum 50); use at least 500 replications for production inference and assess Monte Carlo stability. The principal significance flag and red/gray figures use the pointwise raw test. At $\alpha = 0.05$, significance matches whether the pointwise 95% confidence interval excludes zero. Benjamini-Hochberg fields remain sensitivity outputs and should support any statement about how many domains changed.

10. Understand the model selection rule

MFH2 is the default MFH model. Selecting MFH3 starts the Molina-Romero sequence: an estimation error or non-converged fit moves automatically to MFH2, with the two cases logged separately. A converged MFH3 fit is retained only when its random-effect variance-homogeneity test supports heterogeneity after the selected multiplicity adjustment. MFH1 remains a manual sensitivity model. The optional MFH3 sensitivity fit does not change the selected model; the final report records the model used.

11. Interpret covariates and fallbacks correctly

With LASSO enabled, year-specific UFH or MFH covariate lists define the screening candidate pool. With LASSO disabled, they define the fixed specification. If a boundary MFH fit cannot be benchmarked, Comparison records the reason in **benchmark_status.csv** and continues with unbenchmarked estimates.

12. Treat AI as a separate external transfer

AI is off by default. Enter a provider key only after selecting the external-transfer acknowledgement and confirming that the provider or gateway is institutionally approved. The application does not send household microdata. Prompts include unpublished aggregate estimates, uncertainty measures, diagnostics, counts, and analysis context; geographic identifiers are removed from top-ranked and extreme-change prompt tables. Aggregate unpublished estimates may still be confidential. AI commentary is advisory and requires documented human review before dissemination.

Keep the evidence and complete review

RECORDKEEPING AND PRODUCTION RELEASE

13. Read both reports and keep the evidence

After a successful run, open **outputs/final_report.html** in a browser or **outputs/final_report.docx** in Word. Word text and tables are editable, and figures are embedded images. Save a separate working copy before editing; later runs replace current outputs. Both formats contain the same statistical results and labeled AI text when requested. Keep **outputs/data/** beside the reports for linked Excel workbooks; archived copies are under **app_runs/<timestamp>_<run_label>/outputs/**.

The clean ZIP excludes generated reports. Signed-change figures are under **outputs/figures/change_comparisons/**, with source values in **outputs/data/change_estimate_comparison.xlsx** and **EU_SAE_results.xlsx**. They compare estimated changes in percentage points, not confidence-interval widths. A tighter spread across domain estimates alone does not establish smaller MSE or a statistically significant variance reduction. All maps use one colour scheme. In the final report, maps of one kind (poverty rates, RMSE, poverty change) share one legend range, so colours can be compared across methods and years; maps on the Spain boundary carry the IGN credit.

If report conversion fails, the HTML remains. Check the run log (it names the Pandoc folder used, or why none was available), xml2, and zip, then regenerate after resolving the warning. Retain the exact ZIP, **release_manifest.csv**, verification output, **VERSION**, **WIZARD_VERSION**, run metadata, input hashes, and review approvals. Passing tests or checksums does not approve the statistical methods, data rights, privacy basis, or official publication.

Before a production release. Complete **docs/RELEASE_CHECKLIST.md**, obtain institutional approval for documentation/media rights and release authority, and independently verify the release manifest.